Lecture 21

Monday, April 27, 2026 7:21 PM

Recap: Decision trees for classification ✓

Today: <u>Decision trees for regression</u>, ensemble methods

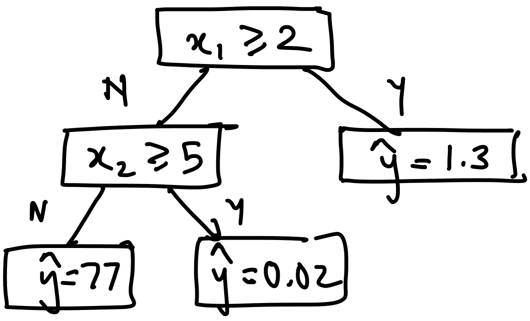
BuildTree: Greedy & recursive algorithm
(Gini impurity index)

Decision trees have high-variance
sensitive to training data

DT classification
Given x, output / predict $\hat{y} \in \{0, 1, \dots K-1\}$

DT regression:
Given x, output $\hat{y} \in \mathbb{R}$

How to use DT for regression? Say $d = 2$



$x = (3, 6)$
$\hat{y} = 1.3$

How to build a decision tree for regression?

SSE: Sum of squared error (TSS)

For a set I, $\bar{y} = \frac{1}{|I|} \sum_{i \in I} y^{(i)}$ ← mean of y values in I

$$SSE(I) = \sum_{i \in I} \left(y^{(i)} - \bar{y}\right)^2$$

mean

square error

sum of square error

SSE very small $\Rightarrow$ $y^{(i)}$s are close to each other
SSE very large $\Rightarrow$ $y^{(i)}$ are far apart from each other

$I \to I^-, I^+$ classification $\downarrow$ GI, good set
$I^- \cup I^+ = I$ regression $\downarrow$ SSE, good set
$I^- \cap I^+ = \phi$

Ensembling
- Use multiple ML models to make one "better" predictor.

Toy example:
- Binary classification
- $\underbrace{\mathbb{P}(\text{error}) = \epsilon}$

Case II
- $\boxed{\text{3 datasets,}}$ 3 models
  $\mathbb{P}(\text{error of any one model}) = \epsilon$
- Models independent of each other

$\underbrace{\qquad\qquad}_{\text{majority voting}}$

Case II

$$\mathbb{P}(2 \text{ models being } x) = \binom{3}{2} \epsilon^2 (1-\epsilon)$$

$$\mathbb{P}(3 \text{ model being } x) = \epsilon^3$$

$$\mathbb{P}(\text{prediction } x) = 3\epsilon^2(1-\epsilon) + \epsilon^3$$

Case ~~II~~ I $\mathbb{P}(\text{prediction } x) = \epsilon$

$$\epsilon > 3\epsilon^2(1-\epsilon) + \epsilon^3$$

$$\Rightarrow \epsilon\left(1 - 3\epsilon(1-\epsilon) - \epsilon^2\right) > 0$$

$$\Rightarrow \epsilon\left(1 - 3\epsilon + 3\epsilon^2 - \epsilon^2\right) > 0$$

$$\Rightarrow \epsilon\left(1 - 3\epsilon + 2\epsilon^2\right) > 0 \qquad (\text{assume } \epsilon < 1/2)$$

$$\Rightarrow \epsilon\,(2\epsilon - 1)\,(\epsilon - 1) > 0$$

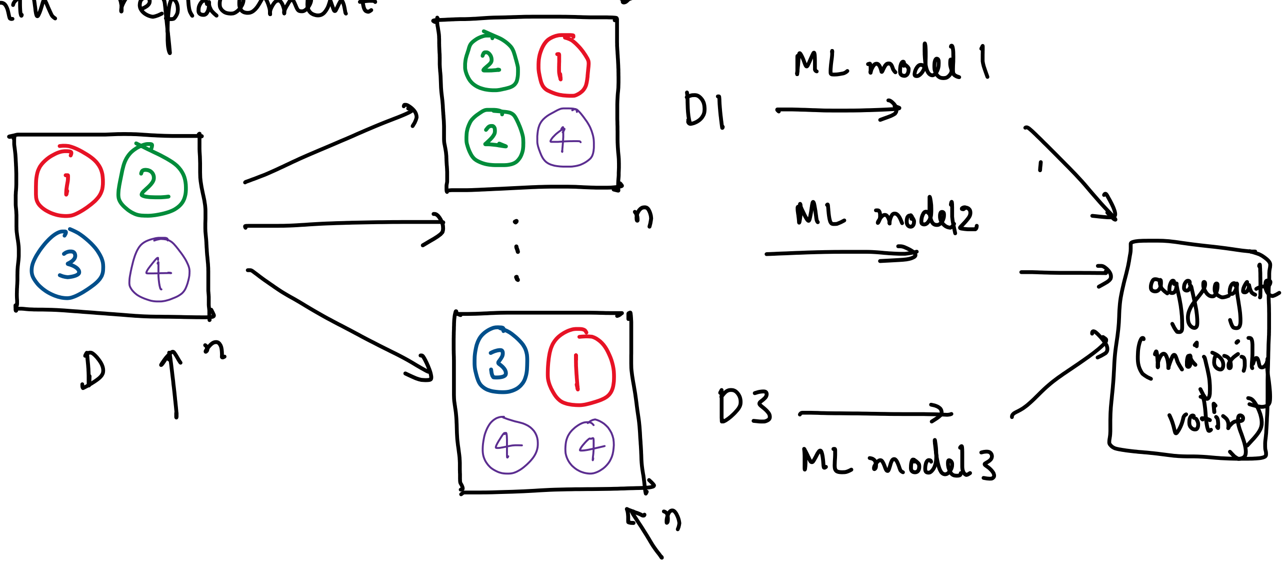
Case I error probability > Case II error probability

$\{x^{(i)}, y^{(i)}\}_{i=1}^{N}$: data set

- We want to train B models, each on "independent data".
- Option 1 : split data into B parts
  - low data, ↑ training error
- Simulate having multiple dataset : (Bootstrapping)

$\{x^{(i)}, y^{(i)}\}_{i=1}^{N}$

- Create B dataset from this by uniformly sampling with replacement



D1 ⟶ decision tree 1

D2 ⟶ decision tree 2 logistic regression

D3 ⟶ decision tree 3 naive bayes

$x \to$ DT1 $\to \hat{y}_1$

$\to$ DT2 $\to \hat{y}_2$

$\to$ DT3 $\to \hat{y}_3$

$\hat{y} = \text{majority}(\hat{y}_1, \hat{y}_2, \hat{y}_3)$

Bagging (Bootstrap aggregating)

1. For $b = 1, 2, \ldots B$
   a) Create dataset by uniformly sampling with replacement

⇒ b) train your favorite ML model. ←

2. Prediction :
   → Classification $\hat{y}$ = majority $(\hat{y}_1, \hat{y}_2, \ldots \hat{y}_B)$
   → Regression $\hat{y}$ = average $(\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_B)$

Random Forest = Bagging + decision trees + extra randomness

1. For $\boxed{b=1}$, 2, ... B
   a) —— " ——
   b) Decision trees
      For every recursive call to Build tree algo
      - pick m out of d features, without replac
      - pick best spilt among only these m featur
2. Prediction using aggregation.

d = 10   m = 2   1, 2, 3, 4, 5, 6, 7, 8, 9, 10